

Titanic Visualization Exercises

Exercise 1 — Line Chart (Beginner)

Create a line chart showing a hypothetical stock price over 30 days. Generate the data using numpy: start at Rs.1000 and add random daily changes (np.random.randn). Plot the price with a red dashed horizontal line at the starting price. Add title, labels, legend and grid.

```
import numpy as np
import matplotlib.pyplot as plt

np.random.seed(42)
days = np.arange(1,31)
daily_changes = np.random.randn(30) * 10
prices = 1000 + np.cumsum(daily_changes)

plt.figure(figsize=(10,5))
plt.plot(days, prices, marker='o', label='Stock Price')
plt.axhline(y=1000, color='red', linestyle='--', label='Starting Price')
plt.title("Hypothetical Stock Price Over 30 Days")
plt.xlabel("Day")
plt.ylabel("Price (Rs.)")
plt.grid(True)
plt.legend()
plt.savefig("stock_line_chart.png", dpi=120)
plt.show()
```

Output: stock_line_chart.png

Exercise 2 — Bar and Horizontal Bar

Using Titanic dataset: 1. Create a bar chart showing average fare by passenger class. 2. Create a horizontal bar chart showing survival rate (%) by embarkation port sorted highest to lowest. Add value labels.

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("titanic_cleaned.csv")

# Average Fare by Passenger Class
avg_fare = df.groupby("Pclass")["Fare"].mean()

plt.figure(figsize=(7,5))
bars = plt.bar(avg_fare.index.astype(str), avg_fare.values)

for bar in bars:
    plt.text(bar.get_x()+bar.get_width()/2,
             bar.get_height(),
             f"{bar.get_height():.2f}",
             ha="center")

plt.title("Average Fare by Passenger Class")
plt.xlabel("Passenger Class")
plt.ylabel("Average Fare")
plt.show()

# Survival Rate by Embarked
survival = (df.groupby("Embarked")["Survived"]
            .mean()*100).sort_values(ascending=False)

plt.figure(figsize=(7,5))
bars = plt.barh(survival.index, survival.values)

for bar in bars:
    plt.text(bar.get_width()+1,
             bar.get_y()+bar.get_height()/2,
             f"{bar.get_width():.1f}%",
```

```

        va="center")

plt.title("Survival Rate by Embarkation Port")
plt.xlabel("Survival Rate (%)")
plt.show()

```

Output: 1) Average Fare Bar Chart 2) Survival Rate Horizontal Bar Chart

Exercise 3 — Histogram Analysis

Create side-by-side histograms for Age and Fare with mean (red) and median (orange).

```

fig, ax = plt.subplots(1,2, figsize=(12,5))

# Age Histogram
ax[0].hist(df["Age"].dropna(), bins=20)
ax[0].axvline(df["Age"].mean(), color="red", label="Mean")
ax[0].axvline(df["Age"].median(), color="orange", label="Median")
ax[0].legend()

# Fare Histogram
ax[1].hist(df["Fare"], bins=20)
ax[1].axvline(df["Fare"].mean(), color="red", label="Mean")
ax[1].axvline(df["Fare"].median(), color="orange", label="Median")
ax[1].legend()

plt.show()

```

Output: Two Histograms (Age and Fare)

Exercise 4 — Seaborn Charts

Create 2x2 dashboard and save as seaborn_dashboard.png.

```

fig, ax = plt.subplots(2,2, figsize=(12,10))
# sns.boxplot(...)
# sns.violinplot(...)
# sns.countplot(...)
# sns.heatmap(...)
plt.savefig("seaborn_dashboard.png")
plt.show()

```

Output: seaborn_dashboard.png

Exercise 5 — Story Chart

Create one insight chart using plt.annotate().

```

# Create your chart
plt.annotate("Key Insight",
            xy=(1,1),
            xytext=(2,2),
            arrowprops=dict(arrowstyle="->"))

plt.show()

```

Output: Story Chart with Annotation

Stretch Challenge

Create a 3x2 Titanic EDA report and save as titanic_eda_report.png.

```

fig, ax = plt.subplots(3,2, figsize=(14,16))
# Build six required charts
plt.savefig("titanic_eda_report.png", dpi=200)
plt.show()

```

Output: titanic_eda_report.png